

Lesson 2 Ros Common Documents and Terminology Description

1. Ros file system components

Ros files consist of Packages and Manifest (package.xml).

Packages: Software packages constitute Ros applications. Each software package contains libraries, executables, scripts and other things created manually.

Manifest (package.xml): The manifest is a description of the related information of software package, which is used to define the dependencies between related meta-information of the package, including the version, maintainer, and license agreement.

2. Common documents

Urdf: description document of robot can be applied in robot simulation software. It usually contains node, texture, collision, etc.

Srv: Described a service. The srv file is stored in the srv directory, containing request and response. And each request can be separated by "-----".

Package xml: describe the attribute of feature packs, including name, version, author, etc.

Strack.xml: Strack refers to the combination of package, which used to describe the information contained in Package.

CmakeLists.txt: compile the configuration file with the help of Cmake

Launch: script, each node and service needed to start the robot

systematically

3. Basic Ros Terminology

① Node: a node is a executable and it can communicate with other nodes through Ros

② Topic: node can receive messages by sending message to topic or subscribing topic.

③ Message: Message is one type of Ros data, which is used to subscribe and be sent to topic.

④ Service: Service is another way to communicate between nodes. The service allows a node to send a request and get a response.

⑤ Master: used for Ros name service

4. Common Tools

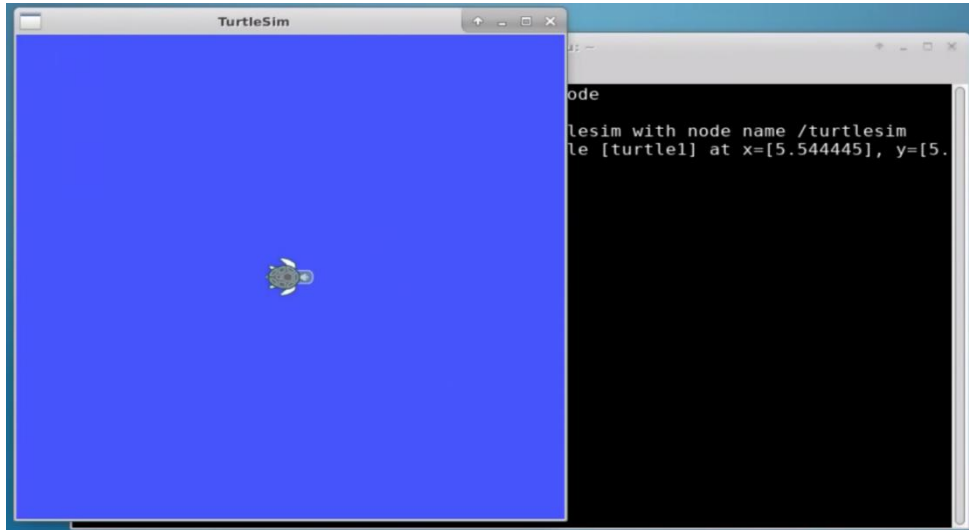
4.1 Turtlesim

Turtlesim is used for Ros or Ros software teaching.

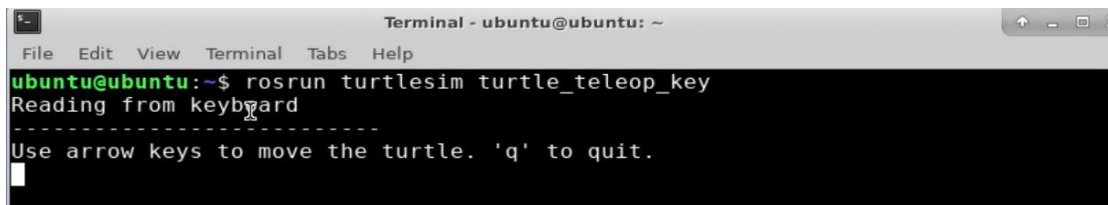
① Before using, please open the command line terminal.

② Enter “roscore” command and open **master**, mentioned before, which is used to manage nodes.

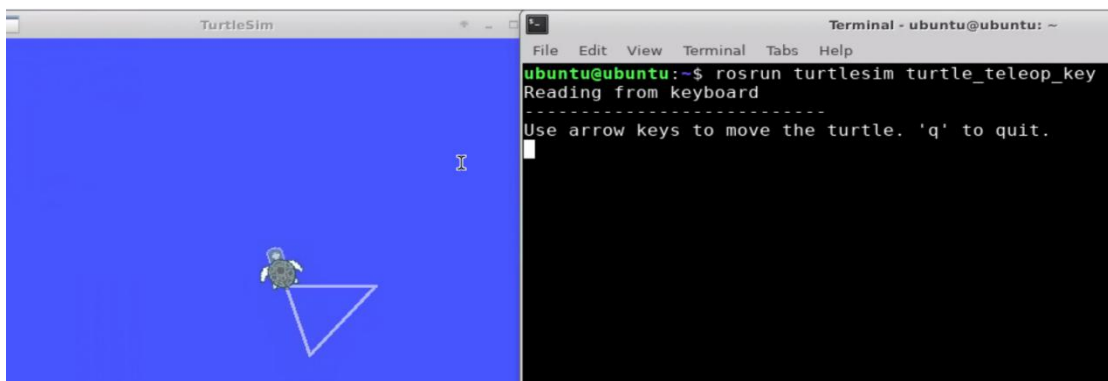
③ After opening master, we need to enter “**roslaunch turtlesim turtlesim_node**” command to start Turtlesim. And now, we open a node also called a process.



④ Open a new terminal again and enter “**roslaunch turtlesim turtlesim_teleop_key**” command. Control the movement of the tortoise with front, back, right and left key.

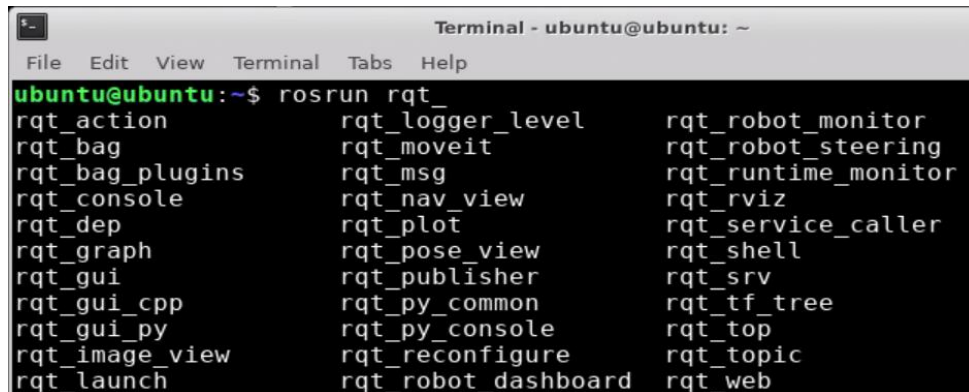


Warm notice: pin top the terminal interface to control.



4.2 QT visual tool

Open terminal and enter “**roslaunch rqt**” command. Then press “Tab” key to unfold the following commands.

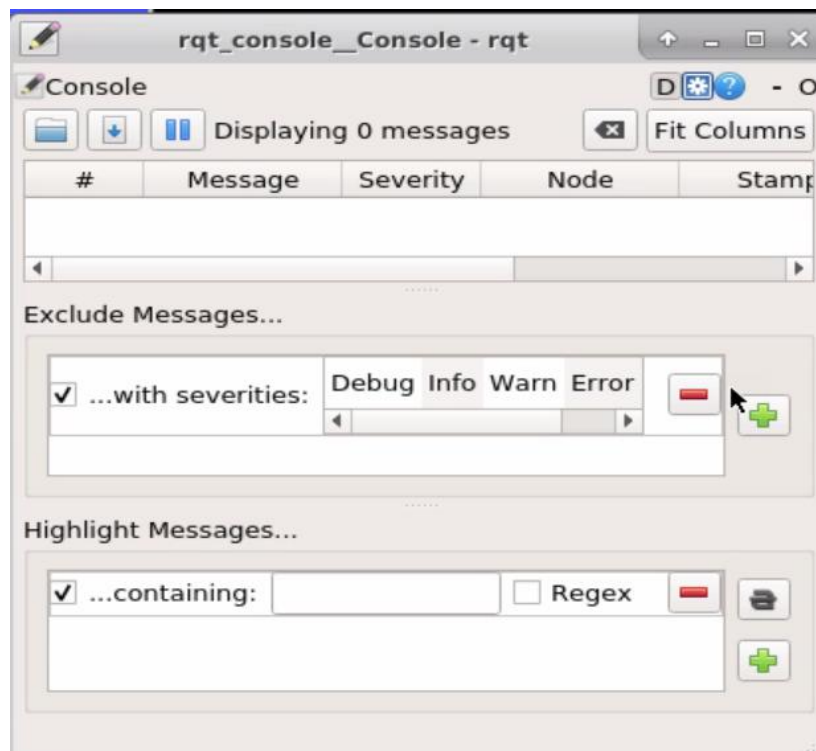


```

Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch rqt
rqt_action          rqt_logger_level   rqt_robot_monitor
rqt_bag             rqt_moveit         rqt_robot_steering
rqt_bag_plugins     rqt_msg            rqt_runtime_monitor
rqt_console         rqt_nav_view       rqt_rviz
rqt_dep            rqt_plot           rqt_service_caller
rqt_graph           rqt_pose_view      rqt_shell
rqt_gui             rqt_publisher      rqt_srv
rqt_gui_cpp         rqt_py_common      rqt_tf_tree
rqt_gui_py          rqt_py_console     rqt_top
rqt_image_view      rqt_reconfigure    rqt_topic
rqt_launch          rqt_robot_dashboard rqt_web
  
```

In “3.AI Vision Games Lesson”, we have used **rqt_image_view** to turn on image returned by camera. And here, we introduce other tools to you.

- 1) **rqt_console**: toolkit for outputting log

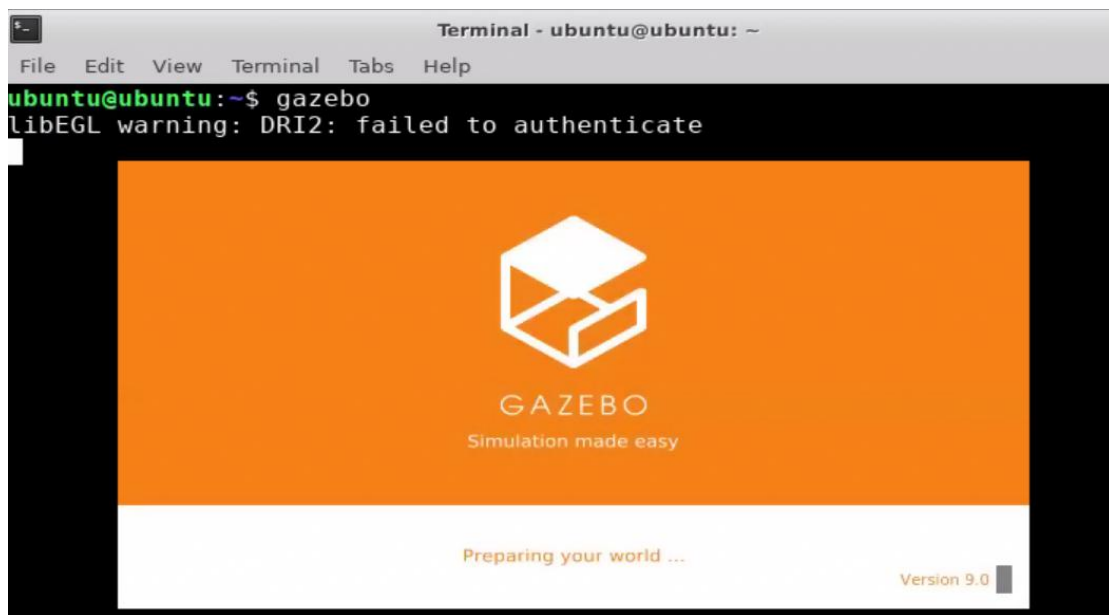


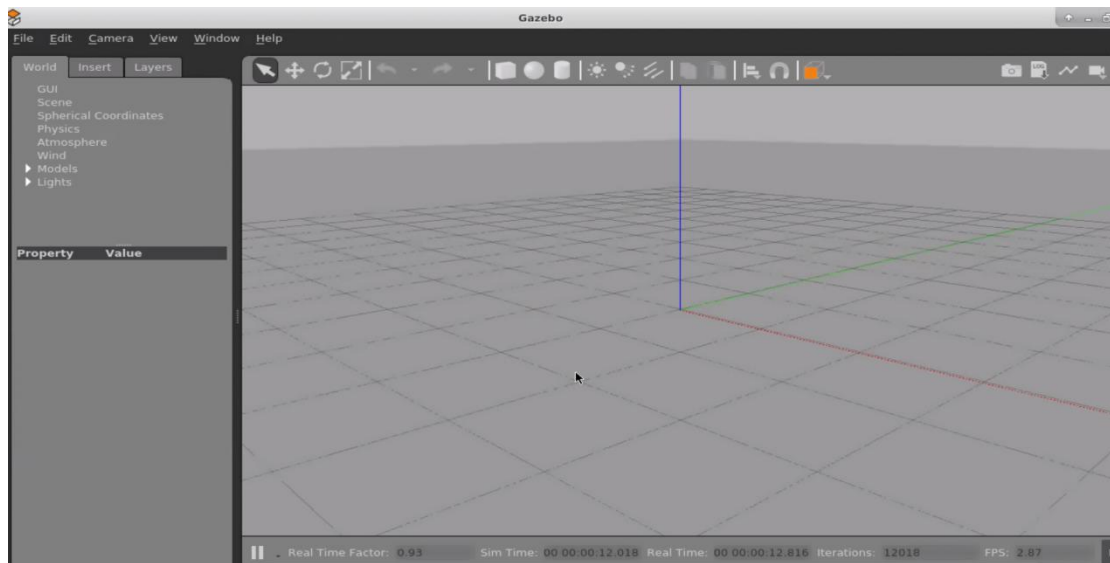
- 2) rqt_graph: visual tool for calculation chart
- 3) rqt_plot: visual tool for calculation chart
- 4) rqt_reconfigure: configuration tool for parameter dynamics

4.3 Gazebo Physical Simulation Environment

Through building various models, it can make virtual robot work in a real scenario.

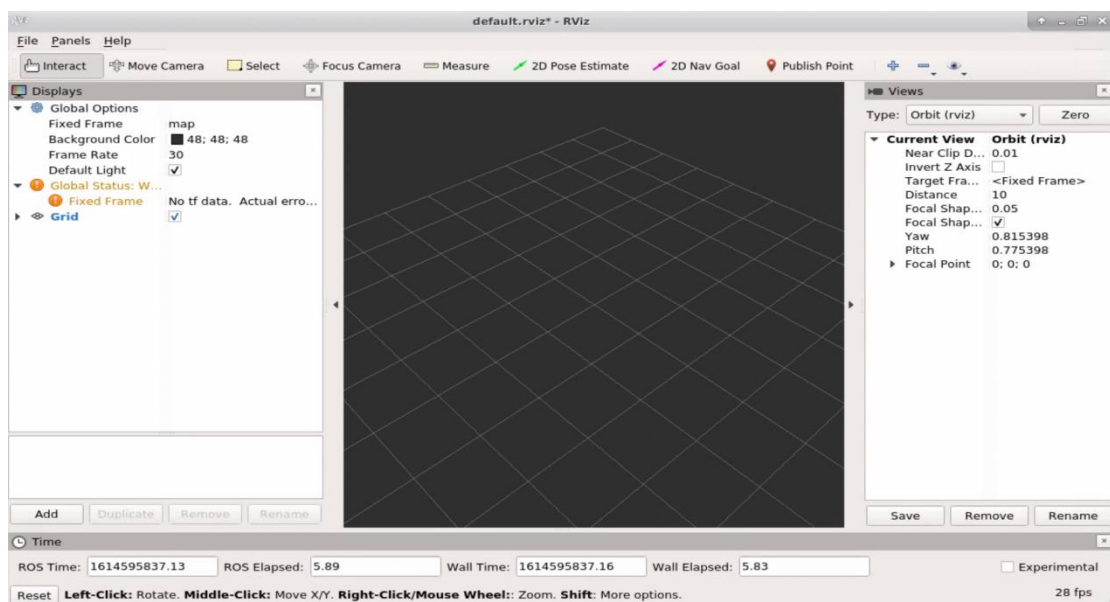
Enter “gazebo” command and wait to open the gazebo.





4.4 Rviz visual platform

It is used to simulate a variety of data of robot, and can be combined with suitable plug-ins.



5. Common Command

Through previous explanation, we have a understanding of basic Ros command. And other basic commands are listed below.

Name	Meaning
roscore	Combination of nodes and programs and can be used to open Ros service.
roscd	Switch the path.
rosls [package_name]	Check the file in software package directory.
rospack find [package name]	Search through package name.
rospack depends [package_name]	Check the libraries relied by package
roscd list	List the running nodes.
rostopic list	Obtain the current active topic.
rosservice list	Display service information of activity.
rosservice call [service] [args]	Call for service with the input parameter
roscd [package_name] [filename]	Edit the file locating in some package through default editor.

<code>roslaunch [package_name] [node_name]</code> <code>[__name:=new_name]</code>	Run the node.
<code>roslaunch show message-type-name</code>	View the details of message type.